

RAKAN ALRASHEED

AI Engineer — LLMs · Agentic Systems · Computer Vision

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PROFILE

AI Engineer specializing in **large language models, agentic / multi-agent systems, and computer vision**, with a track record of taking production AI from concept to deployment across enterprise and startup settings. Designed and shipped an open-source AI workforce platform now used by **many businesses**, and delivered AI-driven automation that raised team productivity **~60%** on Saudi Arabia's national **Tawakkalna** platform at SDAIA. Strong across the full stack — Python / TypeScript services, self-hosted LLM infrastructure, RAG, and computer vision — with a consistent focus on measurable business impact.

CORE SKILLS

AI & Machine Learning: LLMs, Agentic & Multi-Agent Systems, RAG, Prompt Engineering, Computer Vision, YOLO, CNNs, NLP, PyTorch (torch), Deep Learning, Reinforcement Learning (foundational)

Programming: Java, Python, C, C++, Swift, TypeScript, JavaScript, HTML, CSS, SQL, PHP

Web & Backend: React, Next.js, FastAPI, Node.js, REST APIs, Laravel

Cloud, Tools & DevOps: AWS, Azure, Docker, Kubernetes, Git, GitHub, Linux, CI/CD, DevOps, Agile, MySQL, NoSQL, PostgreSQL, Redis, Weaviate, MCP, vLLM, llama.cpp, OpenHands, Headlamp, Argo CD

Testing & QA: Robot Framework, Selenium, Pytest, JUnit, Locust, JMeter (ISTQB Certified)

Soft Skills: Public Speaking, Collaboration, Problem-Solving, Adaptability, Communication, Fast Learner, Scrum

Languages: Arabic (native), English (fluent)

PROFESSIONAL EXPERIENCE

AI Automation Engineer | SDAIA, Riyadh

Mar 2024 – Present

- Accelerated delivery on **Tawakkalna** — Saudi Arabia's national super-app serving tens of millions of users — by building **Genexa**, an AI platform that auto-generates manual, UI, and API test cases and converts business requirement documents (BRDs) into user stories for product owners.
- Standardized testing across teams by architecting the core **API and UI test-automation infrastructure** in **Robot Framework** with reusable, shared frameworks.
- Lifted colleagues' productivity **~60%** by running hands-on **workshops** that drove team-wide adoption of the automation infrastructure and Genexa platform, embedding secure on-machine AI into daily enterprise workflows.
- Simplified performance-test authoring by designing a layer that integrates **Locust with Robot Framework** — work recognized with an **invitation to present at a conference in Finland**.
- Strengthened test reliability by integrating a **Retrieval-Augmented Generation (RAG)** model that powers AI-driven test-case generation and validation.
- Enabled precise UI verification by applying **computer vision** for locator detection and visual-component validation.
- Improved software reliability and performance by engineering automated testing with **Robot Framework, Python, and AI agents**.

Data Consultant | Devoteam, Riyadh

Oct 2023 – Mar 2024

- Strengthened security and surveillance for the **Ministry of Interior** by developing a **computer-vision motion-detection system**.
- Safeguarded data integrity and governance by building **data pipelines** with **Informatica Axon, EDC, and DQ**.
- Improved data maturity across multiple domains by supporting **HRSD's** alignment with **NDI** compliance standards.
- Automated customer inquiries by developing an **AI-powered chatbot** integrated with phone and messaging platforms.

Software Engineer (Intern) | Ecorithms, San Diego, CA

Jul – Sep 2023

- Delivered a mobile web application for the solid-waste industry using **Svelte** (front-end), **Node.js** (back-end), and **MongoDB** for data management.
- Reached **89% accuracy** auditing, recognizing, and categorizing waste by integrating extensive datasets with **YOLO** real-time object detection.
- Improved test coverage **35%** by automating test cases with **Selenium and Pytest**, running continuous post-sprint testing, and using **JMeter** for performance and load testing in an Agile environment.
- Ensured smooth deployment and continuous feedback by setting up **Jenkins** for CI and automated testing.
- Maintained compliance with data-protection best practices by automating data-handling systems.

Web Developer (Intern) | InterOcean Systems, LLC, San Diego, CA

Jun – Aug 2022

- Increased website traffic **40%** by designing, developing, and maintaining responsive, user-friendly sites in **HTML, CSS, TypeScript, and React**.
- Validated the scalability and responsiveness of web applications by implementing performance tests.
- Hardened applications by conducting **OWASP ZAP** security testing — identifying vulnerabilities and establishing data-privacy policies with the team.

PROJECTS

Navaia — AI Workforce Platform | Founder & Lead AI Engineer

Oct 2024 – May2026

- Built and shipped **NavaiaForge**, an open-source, self-hostable multi-agent AI “workforce” platform now deployed by **22 businesses and cutting task-handling time 40–60%** — specialized agents share context, route work through a **visual canvas builder**, ground answers in **RAG knowledge bases (Weaviate)**, and stream results in real time.
- Architected the full stack — Next.js canvas UI, **FastAPI** backend, PostgreSQL / Redis, and a dual agent runtime (Claude + self-hosted open-weight models) — with built-in **observability, cost tracking, and RL-based agent evaluations**.
- Published **Python & TypeScript SDKs** (PyPI + npm) over the platform, enabling fully code-driven workforce creation, task orchestration, and real-time WebSocket streaming.
- Built a **Coding agent**, an open-source, LLM-agnostic AI coding agent in pure Python (an alternative to OpenHands / Claude Code) — 50+ tools, concurrent streaming tool use, **token-budget-aware memory compaction with context restore**, and a fine-grained permission model.
- Enabled regulated customers to keep all data on their own infrastructure by designing for **on-prem / air-gapped deployment** (single Docker command); unified voice, chat, and HR-recruiting agents as coordinated roles within one workforce.

Obstacle Avoidance App | Project Lead

Aug 2023 – May 2024

- Won the **Computer Science Best Capstone Award** for an AI navigation aid that helps visually impaired users avoid obstacles through real-time audio feedback from a chest-worn phone.
- Led the application coding (**Swift**) and **AI integration**, training the model on custom datasets with **YOLOv8** for precise obstacle detection and distance calculation.
- Ensured seamless functionality at each iteration through sprint-based development with unit and **BDD** testing, in collaboration with the team.

EDUCATION

B.S. in Computer Science — Shiley-Marcos School of Engineering, University of San Diego, CA

Dec 2023

- Concentration in Artificial Intelligence & Data Science · Dean’s List Academic Achievement Award.
- **Relevant Coursework:** Data Structures, Algorithms & Analysis, Advanced Computational Problem Modeling, Automata & Computability, Data Science Foundations, Intro to AI, Object-Oriented Design, Web Programming, Neural Networks.

Artificial Intelligence & Machine Learning — Harvard University (Certificate)*Jun 2024*

AWARDS & CERTIFICATIONS

- **NVIDIA-Certified Professional Agentic AI (NCP-AAI)** · May 2026
- **Robot Framework Certified Professional (RFC)** · Jan 2026
- **ISTQB Certified Tester – Foundation Level (CTFL)** · Mar 2025
- **Informatica Certified Professional (ICP) – Data Engineering 10.2 Developer** · Dec 2024
- **AI & Data Science Bootcamp – SDAIA** — intensive program in RL, ML, Data Science, NLP, and LLMs applied to real-world AI projects · Jul–Oct 2024
- **Alcalá 100** — selected to the top 100 of USD’s Class of 2024 for exceptional leadership and academic excellence · May 2024
- **ICPC 2023 Regional Programming Contest** — top-ranked among San Diego universities; advanced to the 2024 National Contest.